From Weak Textures to Dense Arrangements: Leveraging Prior Knowledge for Small-Object Detection in Remote Sensing Images

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***Abstract*— Detecting small objects in remote sensing images (RSIs) is a significant challenge due to their weak texture, scale variations, and dense spatial arrangements. Existing approaches often overlook the importance of prior contextual information and the aggregation of features in densely packed small objects, both of which are crucial for improving the performance of remote sensing small-object detection (RSSOD). In this work, we propose the prior-guided context fusion network (PGCFNet), which enhances small-object detection by decoupling scene contextual information through three novel components: the prior-guided context fusion module (PGCFM), the depthwise aggregator (DWA), and the prior-guided small-object detector (PGSOD). This architecture facilitates a deeper exploration of the relationships between small objects and their surrounding environment. Specifically, PGCFM improves feature representation by integrating multiscale features and applying prior-guided dynamic channel weighting, addressing the challenge of weak textures. Additionally, DWA refines feature aggregation using dilated convolutions and dynamic feature adjustment, enabling precise multiscale detection in environments with dense small objects. Furthermore, PGSOD leverages prior knowledge to reduce background interference, enhancing small-object detection across varying scales and orientations. Collectively, these modules work synergistically to advance small-object detection in RSIs, overcoming key challenges in complex environments. Extensive experiments on three public datasets demonstrate that the performance of the proposed method outperforms several state-of-the-art (SOTA) detectors, especially for tiny object detection. Specifically, PGCFNet achieves 86.0% mean average precision (mAP) on the DIOR dataset, 95.59% mAP on the NWPU VHR-10 dataset, and 58.5% mAP on the AI-TOD dataset. Additionally, we conducted generalization experiments for PGCFM, DWA, and PGSOD, demonstrating its effectiveness across different datasets and detection networks with varying model sizes.**

***Index Terms*— Prior acknowledge, remote sensing image (RSI), small-object detection, texture and boundary information.**

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I. Introduction

REMOTE sensing object detection (RSOD) aims to identify and localize objects of interest in remote sensing images (RSIs). This task not only involves recognizing the object category but also accurately determining its position, making it a particularly challenging problem. It plays a crucial role in various applications, including environmental monitoring [1], [2], disaster prediction [3], [4], traffic supervision [5], [6], and so on. In recent years, RSOD has attracted increased attention due to advancements in computing resources and continuous improvements in deep neural network models. These developments have led to significant enhancements in the performance of existing object detectors, particularly for natural images. However, due to the inherent limitations of RSIs in terms of resolution and quality, interested objects are often characterized by weak texture, scale variability, and dense arrangement, as shown in Fig. 1. These factors cause aliasing between objects and backgrounds, making small objects indistinguishable.

To alleviate these challenges, many existing methods have largely focused on feature extraction and background suppression. For instance, Wu et al. [7] updated U-Net to enhance the distinguishability of infrared small objects by improving its ability to extract both local and global features. Gao et al. [8] put forward a local attention pyramid module to highlight small-object features and suppress background noise through patch processing in the channel dimension. Ma et al. [9] identified scale confusion in shallow features as a factor impacting small-object performance and attempted to eliminate large object features in the lower layers. Liang et al. [10] proposed MidNet, an anchor-and-angle-free detector for oriented ship detection in aerial images. It predicts the center and four midpoints of objects, enhancing features via symmetrical deformable convolution and refining keypoints through analytical geometry. Feng et al. [11] introduced SDANet, a density adaptive network for moving vehicle detection in satellite videos. It predicts cluster proposals and centers

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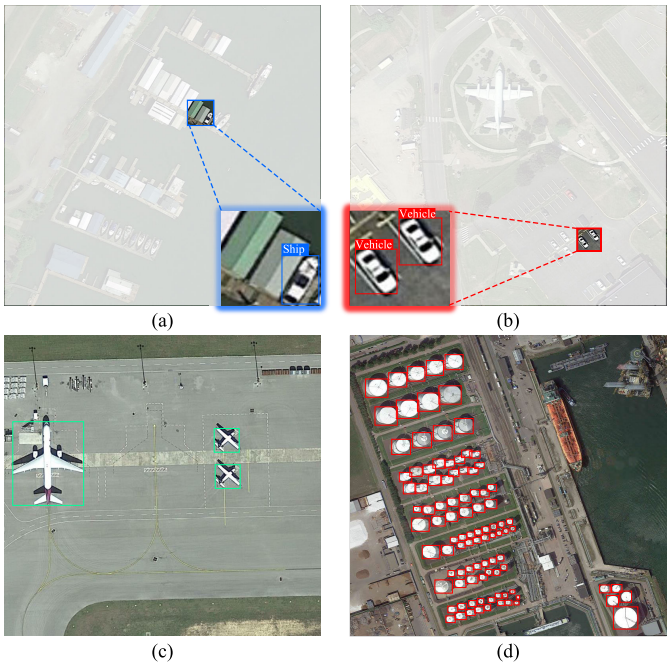


Fig. 1. Properties of RSIs. (a) and (b) Difficulty in classification posed by weak texture. (c) Complex scale variation. (d) Dense object arrangement.

through pixel-wise prediction, matches them using a density-matching algorithm, and embeds road semantics to reduce false detections.

Despite these advancements, the methods mentioned above overlook a key limitation of deep networks: the nonlearnable nature of downsampling and upsampling. Downsampling reduces the resolution of feature maps, causing the loss of weak texture and boundary information. This makes it harder for subsequent layers to capture and differentiate these details. On the other hand, upsampling, lacking learning capability, amplifies background noise indiscriminately. As a result, it becomes more challenging to distinguish objects from the background, potentially obscuring small objects within the noise. As shown in Fig. 2, object detection becomes more challenging after multiple downsampling and upsampling. The region of interest (ROI) shifts away from the actual objects, often aligning with background textures or noise. This misalignment severely affects detection accuracy, particularly in the case of small-object detection in RSIs.

To address the mentioned issues, a novel prior-guided context fusion network (PGCFNet) is proposed. This network dynamically integrates texture, boundary, and background information as prior knowledge, enhancing the model's localization and classification capabilities. PGCFNet leverages the fact that prior knowledge can effectively handle the inherent limitation of RSIs while compensating for information loss and noise accumulation caused by downsampling and upsampling. This is particularly beneficial for small objects with weak textures or limited features, as it provides contextual support to prevent the object from being overwhelmed by the complex background. Specifically, PGCFNet consists of three innovative plug-and-play modules: the prior-guided context fusion module (PGCFM), the depthwise aggregator (DWA), and the prior-guided small-object detector (PGSOD). PGCFM is

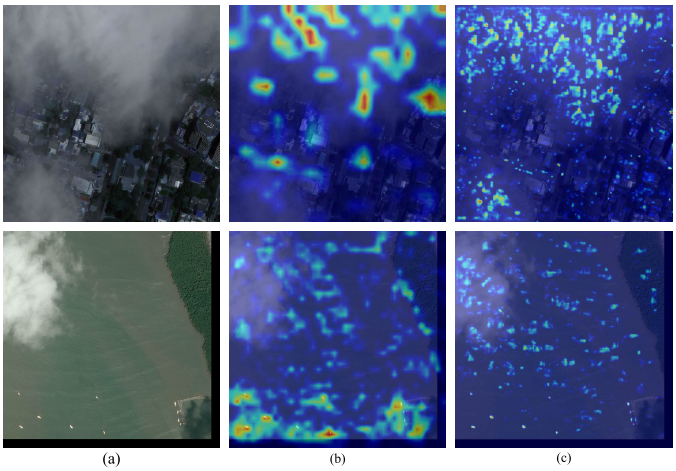


Fig. 2. Illustration of misclassification and false localization in RSOD. (a) Original RSIs. (b) Feature heatmap for classification. (c) Feature heatmap for localization.

designed to handle weak textures in small objects by utilizing multiscale feature extraction and a prior-guided dynamic channel reweighting mechanism. By integrating spatial and contextual information, PGCFM enhances feature separability and reduces false alarms. The mechanism also leverages dynamic prior knowledge of object textures and backgrounds, improving detection precision. In addition, DWA addresses the issue of dense object arrangements by expanding the context and aggregating multiscale features. Its dynamic feature optimization ensures robust discrimination between tightly packed objects and the background, enabling more accurate detection in cluttered scenes. Finally, PGSOD is tailored for detecting small objects with varying orientations and scales. It combines a prior-guided dynamic channel reweighting mechanism with advanced feature extraction strategies, ensuring precise detection under various spatial transformations. Together, these modules form a unified detection pipeline that significantly enhances the ability to detect small objects with complex characteristics in remote sensing environments. Extensive experiments on challenging remote sensing datasets, including DIOR [12], NWPU VHR-10 [13], RSOD [14], and AI-TOD [15] (many instances meet the criteria for small weak object detection [16]), clearly demonstrate the superiority of PGCFNet. The main contributions of this research are as follows.

1) We introduce PGCFM, which addresses weak texture issues by combining multiscale feature extraction with a prior-guided dynamic channel reweighting mechanism. This improves feature separability and reduces false alarms.
2) We present DWA, which mitigates the challenges posed by dense object arrangements by expanding the context and aggregating multiscale features. This optimization enhances the differentiation between tightly packed small objects and the background.
3) We propose PGSOD, which detects small objects across varying orientations and scales, integrating prior-guided context fusion with advanced feature extraction strategies to ensure robust detection under spatial transformations.

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The remainder of this article is organized as follows. Section II reviews related works on texture and boundary information as well as multiscale object detection. Section III describes the proposed PGCFNet architecture. Section IV outlines the experimental setup and results, comparing the proposed method with various benchmark models and several state-of-the-art (SOTA) approaches. Section V concludes the article and discusses future directions for small-object detection in remote sensing.

II. Related Work

A. Texture and Boundary Information

In RSOD, texture and boundary information are crucial for object classification, as they provide essential distinguishing characteristics. Texture features describe pixel distributions with periodic and repetitive patterns within an image, whereas boundary information highlights the edges that separate objects from their background, capturing their shape, contour, and structure. Accurate extraction of boundary information is vital for precise object localization and classification. For example, Li et al. [17] proposed a texture-aware enhancement module (TAEM) and a boundary-aware fusion module (BAFM), which capture interpixel correlations to perceive texture distributions within objects. These modules enhance texture features and extract edge information to guide the detection of small, weak objects. Tu et al. [18] proposed the Edge Guidance Block, which embeds edge prior knowledge into hierarchical feature maps. This approach not only operates at the feature level but also performs spatial-level transformations to effectively integrate edge information. Li et al. [19] developed a hybrid residual network incorporating structure-guided feature transformation to capture distinctive structural characteristics, including edges and corners. Ren et al. [20] proposed constructing a texture representation of objects within the channel dimension and introduced a boundary consistency loss to refine boundary details in camouflaged object detection. Guan et al. [21] introduced an edge detection stream that integrates multiple side outputs via concatenation and employs a fusion layer with a 1×1 convolution to generate a unified output.

Incorporating texture and boundary information has proved effective for RSOD, as these methods assume such features are inherently salient. However, in RSIs, texture and boundary cues are often subtle or even absent and are frequently obscured by complex background patterns, such as vegetation, buildings, or roads. These background elements can generate edge signals that resemble object boundaries, leading to the misidentification of background textures as edges of small objects. This issue ultimately diminishes the accuracy of object detection. In contrast to these methods, we propose a novel approach that dynamically integrates both background and boundary information as prior knowledge to support the detection process. Our method not only improves the robustness of detecting small objects with weak textures but also enriches the contextual understanding, providing a more comprehensive semantic framework.

B. Multiscale Object Detection

The scale and size of objects in RSIs vary due to factors such as their location and the altitude at which they are captured. Research into multiscale feature fusion has shown great potential for improving the accuracy of contextual information aggregation, thereby enhancing the performance of object detectors. Fan et al. [22] introduced a multiscale feature fusion module (MSFFM) that employs an attention mechanism. This module uses matrix multiplication to evaluate the relevance between feature maps in the spatial domain. Adjusting the weights of lower-level feature maps reduces interference across layers. Similarly, Jaderberg et al. [23] proposed a bidirectional feature fusion module (Bi-DFFM) that efficiently aggregates multiscale features through top-down and bottom-up information exchange, enhancing detection performance for objects at different scales. Wang et al. [24] proposed a progressive feature polishing network (PFPN) that utilizes multiple feature polishing modules (FPMs) recurrently. This network detects salient objects with fine details, without requiring postprocessing. Each FPM updates features at its level, integrating higher-level contextual information while preserving the hierarchical structure of the feature maps. Liu et al. [25] proposed a high-resolution object detection network (HRDNet) designed for small vehicle detection, which incorporates a multidepth image pyramid and a multiscale FPN to enhance feature depth. Zhang et al. [26] introduced a novel coarse-to-fine framework (CoF-Net) for RSOD. This framework includes a fine-grained feature alignment (FFA) module that adaptively learns pixel-wise transformations and establishes precise multiscale feature correspondences.

Although these algorithms have enhanced overall detection performance through multiscale object detection, they overlook the spatial correspondence issues inherent in the feature fusion process. This oversight fails mechanisms designed to extract and merge information across scales, leading to the disappearance or submergence of features corresponding to small objects, thus hindering effective recognition. To address this challenge, we propose the DWA for multiscale feature fusion. The DWA dynamically assigns weights to feature maps across scales, effectively mitigating the spatial misalignment caused by nonlearnable upsampling techniques. Furthermore, we introduce the PGSOD, which incorporates prior knowledge into the detection process, enhancing both the robustness and precision of multiscale object detection, particularly for small objects. These proposed innovations significantly improve the integration of multiscale features and demonstrate strong potential for overcoming the unique challenges associated with RSI analysis.

III. Proposed Method

A. Overview

The framework of PGCFNet is summarized in Fig. 3. For the input RSI, multiscale feature maps are first extracted using the backbone network, CSPDarkNet53. These feature maps are then processed by PGCFM, which addresses challenges associated with detecting small objects in RSIs. Such challenges arise due to weak textures, which make accurate

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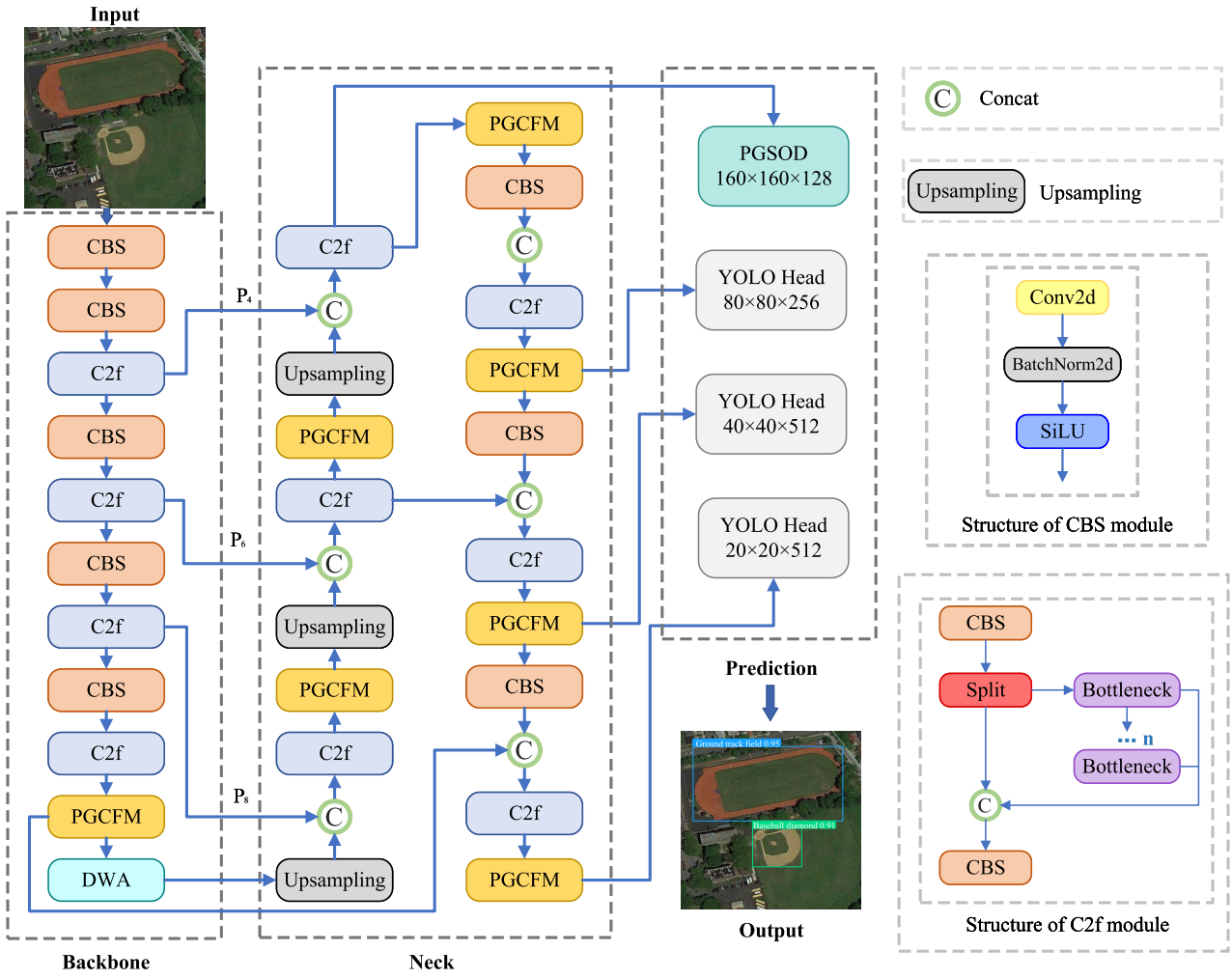


Fig. 3. Overall framework of PGCFNet. PGCFNet architecture takes anchor-free detector YOLOv8 as the baseline, CSPDarkNet53 as the backbone to extract multiscale features from the input RSI, PGCFM, and DWA are combined into the neck and backbone to enhance the detection capabilities of the model, and the detection results are output by the PGSOD and other detection heads.

localization difficult and object boundaries hard to distinguish. To tackle the dense distribution of small objects, DWA applies a dynamic weighting mechanism to enhance and refine the feature maps. The enhanced features are then passed to PGSOD, which integrates the PGCFM with a refined feature extraction strategy to handle small objects across varying orientations and scales. This integration ensures accurate detection, even under spatial transformations. Finally, the processed multiscale feature maps are fed into parallel detection heads for RSOD. This integrated pipeline significantly improves detection performance, particularly for small and complex objects in challenging remote sensing environments.

B. Prior-Guided Context Fusion Module

Small objects in RSIs are often characterized by weak texture, leading to the appearance of similar features across objects of different classes. This phenomenon increases the likelihood of feature confusion in detection models, resulting in a high false alarm rate and reduced detection precision. To address these issues, we propose the PGCFM. This module utilizes multiscale spatial feature extraction and a prior-guided dynamic channel weighting mechanism to enhance the model's discriminative ability for small-object features. By incorporating contextual information and adaptively re-weighting feature maps, PGCFM reduces the risk of false alarms and improves detection accuracy in complex remote sensing scenarios.

PGCFM consists of four main components: multiscale spatial feature extraction, fusion convolution, prior-guided dynamic channel weighting, and a final feature enhancement layer. Section III-B1–III-B4 provides a detailed description of its workflow, and the framework of the proposed PGCFM is illustrated in Fig. 4.

1) Multiscale Spatial Feature Extraction: The presence of background complexity and feature ambiguity often hampers the accurate detection of small objects in RSIs. To address this challenge, multiscale feature extraction is essential for capturing object features at different resolutions. PGCFM employs multiscale spatial feature extraction through convolutions with varying receptive fields. Specifically, for a given input feature map $\mathbf{X} \in \mathbf{R}^{C\times H\times W}$, where C, H, and W denote the number of channels, height, and width, respectively, PGCFM first applies a standard 3×3 convolution ($\text{Conv}_{3\times3}$) to extract local spatial details. This is followed by a 5×5 dilated convolution [27]

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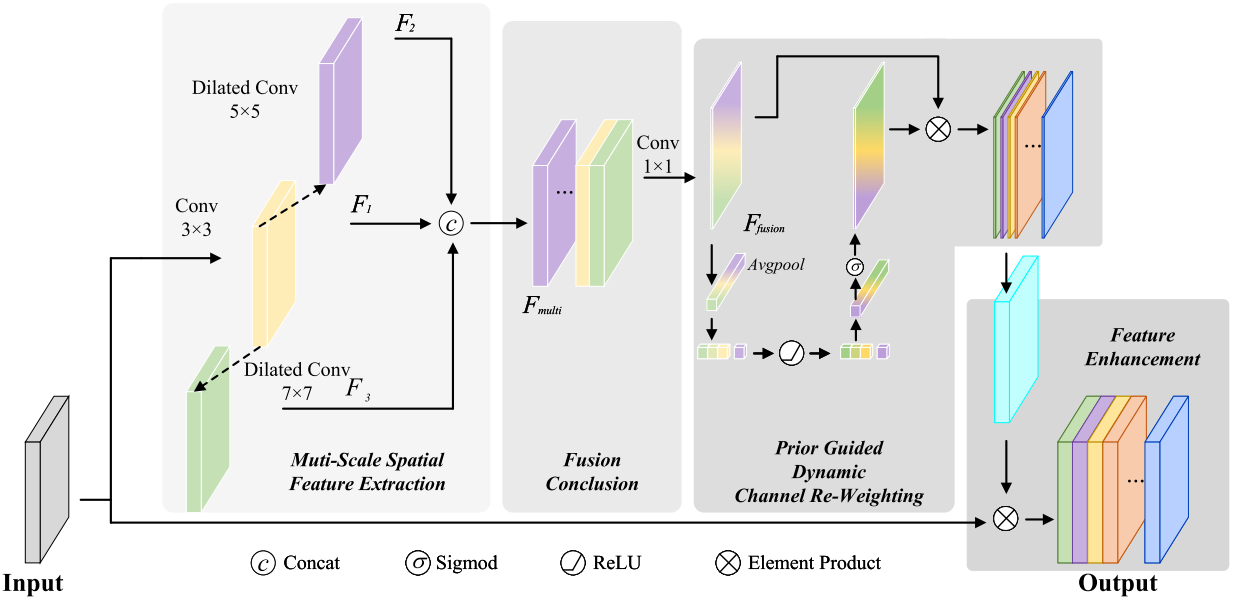


Fig. 4. Structure of PGCFM. The module captures multiscale information using 3 × 3, 5 × 5, and 7 × 7 convolutions and dynamically fuses the resulting feature maps through operations such as convolution and average pooling, effectively highlighting object features.

(DilatedConv$_{5\times5}$) to capture mid-range dependencies and a 7 × 7 dilated convolution (DilatedConv$_{7\times7}$) to incorporate broader contextual information. These operations are formulated as

$$\mathbf{F}_1 = \text{Conv}_{3\times3}(\mathbf{X}) \tag{1}$$

$$\mathbf{F}_2 = \text{DilatedConv}_{5\times5}(\mathbf{F}_1) \tag{2}$$

$$\mathbf{F}_3 = \text{DilatedConv}_{7\times7}(\mathbf{F}_1). \tag{3}$$

The outputs $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3$ correspond to feature maps at different scales. The resulting feature maps are then concatenated along the channel dimension to form a multiscale feature representation

$$\mathbf{F}_{\text{multi}} = \text{Concat}([\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3]). \tag{4}$$

By fusing spatial features across scales, PGCFM enhances the PGCFNet's ability to differentiate between objects with overlapping characteristics.

2) Fusion Convolution: The concatenated multiscale features, denoted as $\mathbf{F}_{\text{multi}}$, have a channel dimension of $3C$, which is computationally expensive and redundant. To address this, a 1 × 1 convolution is applied to reduce the channel dimension back to C

$$\mathbf{F}_{\text{fusion}} = \text{Conv}\,1\times1(\mathbf{F}_{\text{multi}}). \tag{5}$$

This operation effectively integrates the multiscale features while reducing computational overhead, ensuring compatibility with subsequent processing stages.

3) Prior-Guided Dynamic Channel Re-Weighting: To address the issue of false alarms arising from feature confusion in RSOD, PGCFM introduces a dynamic channel re-weighting mechanism. This mechanism aims to emphasize discriminative features while simultaneously suppressing irrelevant or noisy ones. Unlike traditional methods, PGCFM utilizes global prior information, which is adaptively extracted from the feature map itself, to guide the weighting process. The incorporation of global context enhances the model's ability to capture spatially and contextually relevant features, thereby reducing misclassification due to overlapping or ambiguous feature representations. Additionally, PGCFM improves the feature extraction process by incorporating both local and global cues, resulting in a more robust and accurate feature representation. This approach is particularly effective in remote sensing imagery, where fine-grained discrimination between similar objects is critical.

PGCFM employs a prior-guided dynamic channel re-weighting mechanism, enhancing discriminative features while suppressing irrelevant or noisy information. First, global statistics are extracted using a global average pooling operation

$$P = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \mathbf{F}_{\text{fusion}}(i, j). \tag{6}$$

The pooled descriptor $P \in \mathbf{R}^C$ represents global contextual priors. This descriptor is then passed through a gating mechanism to dynamically compute channel weights

$$w = \sigma(\mathbf{w}_2 \cdot \delta(\mathbf{w}_1 \cdot P)) \tag{7}$$

where $\mathbf{w}_1$, $\mathbf{w}_2 \in \mathbf{R}^{C\times(G/r)}$ are learnable weight matrices. The parameter r is the channel reduction ratio, which controls the complexity and computational cost of the module, and σ denotes the sigmoid activation function, and δ denotes the ReLU activation function.

4) Feature Enhancement: Finally, the computed weights, denoted as $w \in \mathbf{R}^C$, are used to reweight the fused features

$$\mathbf{F}_{\text{output}}(c, i, j) = \mathbf{F}_{\text{fusion}}(c, i, j) \cdot w \tag{8}$$

where c indexes the channel and (i, j) represents the spatial location. This operation is designed to highlight channels essential for detecting small objects, which are often difficult

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to identify due to their size and subtle features. Simultaneously, it suppresses irrelevant or redundant features that may introduce noise or lead to false detections. By dynamically adjusting channel importance using spatial and contextual cues, this method ensures that the model prioritizes the most informative channels, thus enhancing its sensitivity to subtle but distinct patterns.

PGCFM enhances feature representation in RSOD by integrating multiscale spatial features and applying prior-guided channel reweighting. It effectively balances local and global contextual information, improving feature expressiveness for detecting small objects with subtle and complex characteristics. By dynamically suppressing ambiguous features caused by low texture and contrast, PGCFM reduces false alarms and increases classification accuracy. Additionally, it improves the separability of objects and background features, facilitating more accurate detection in challenging environments. Consequently, PGCFM significantly boosts detection precision, making it a valuable tool for tackling challenges like small objects, low contrast, and background clutter in RSOD.

C. DepthWise Aggregator

The frequent occurrence of small objects in RSIs often results in significant feature overlap, making it challenging to distinguish objects from the background. This overlap leads to confusion in feature representation due to the interference of neighboring objects during extraction. Consequently, the detection model struggles to capture fine-grained details of small objects, ultimately reducing detection accuracy. YOLOv8 [28] incorporates the spatial pyramid pooling-fast (SPPF) module to enhance the network's receptive field through multiscale pooling. However, this method has limitations in certain scenarios. The SPPF module relies on a straightforward feature concatenation approach, which may fail to fully exploit global contextual information. Additionally, its fixed-scale max-pooling lacks an effective feature selection mechanism, resulting in spatial information loss. This further impairs the network's ability to differentiate densely arranged small objects.

To address this challenge, we propose DWA to replace the original SPPF module, which optimizes feature expression through three key mechanisms: context expansion, dynamic feature optimization, and multiscale feature aggregation, as shown in Fig. 5. These mechanisms significantly enhance the feature expression capability for small objects while suppressing feature interference.

1) Context Expansion: Dense spatial arrangements in RSIs cause interference between neighboring features of small objects, complicating their separation. To mitigate this issue, DWA employs dilated convolution to expand the receptive field, allowing the model to capture broader contextual information while preserving spatial resolution—an essential aspect for separating overlapping features.

The first stage of the DWA module involves context expansion to capture more spatial information. Given an input feature map $\mathbf{X} \in \mathbf{R}^{C\times H\times W}$, a 1×1 convolution is applied to reduce the channel dimension while maintaining the spatial dimensions

$$X_1 = \phi(X) = \mathrm{Conv}_{1\times 1}(X) \tag{9}$$

where $\phi(X)$ denotes the 1×1 convolution operation. Following this, a dilated convolution is applied to further expand the receptive field, capturing multiscale contextual information. The dilated convolution can be expressed as follows:

$$X_2 = \varphi(X_1) = \mathrm{Conv}_{3\times 3,d=2}(X_1) \tag{10}$$

where $\varphi(X_1)$ represents the dilated convolution operation with a dilation rate of $d = 2$. The advantage of using dilated convolution lies in its ability to significantly enlarge the receptive field without increasing the number of parameters. This enables the network to effectively capture distant spatial relationships, which is particularly beneficial for remote sensing small-object detection (RSSOD) tasks.

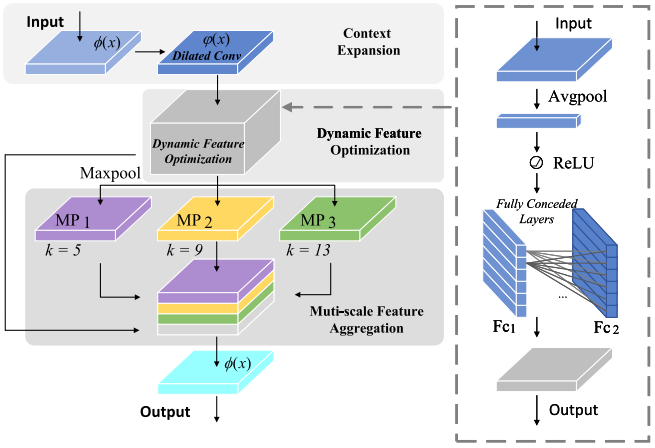


Fig. 5. Structure of DWA. The DWA effectively expands the receptive field using dilated convolutions, enabling features to capture richer contextual relationships. It generates multiscale information through diverse pooling kernels, ensuring comprehensive coverage of both fine-grained and coarse-grained features.

2) Dynamic Feature Optimization: While context expansion and multiscale aggregation offer significant advantages, the presence of complex backgrounds in RSIs introduces noise, further complicating feature discrimination. To address this issue, DWA incorporates a dynamic channel weighting mechanism that adaptively emphasizes salient features while suppressing irrelevant or noisy ones. By modeling global information, this mechanism guides the network to adaptively adjust the feature map responses across different regions.

First, an average pooling (AvgPool) operation is applied to the intermediate feature map X_2 to generate a global descriptor

$$z = \mathrm{AvgPool}(X_2) = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j=1}^{W} X_2(i, j) \tag{11}$$

where z denotes the global context vector and $H \times W$ represents the spatial dimensions. Next, a series of nonlinear transformations are applied to generate dynamic adjustment coefficients

$$s = \mathrm{FC}_2 \cdot \delta(\mathrm{FC}_1 \cdot z) \tag{12}$$

where $\mathrm{FC}_1 \in \mathbf{R}^{(C_1/r)\times C_1}$ and $\mathrm{FC}_2 \in \mathbf{R}^{C_1\times(C_1/r)}$ are fully connected layers and s represents a vector of weights

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consistent with the number of input channels for channel attention adjustment.

Lastly, the dynamically adjusted feature representation is obtained by performing channel-wise scaling

$$X_3 = s \cdot X_2 \tag{13}$$

where X_3 represents the refined feature map, and the scaling operation is performed in a channel-wise manner. This dynamic feature optimization mechanism effectively suppresses irrelevant background information while enhancing the feature response of the object region, which helps to improve the detection accuracy of small objects in RSIs.

3) Multiscale Feature Aggregation: In addition to the issue of dense feature distribution, the size variation of small objects presents a significant challenge in RSOD. Single-scale feature extraction methods are inherently limited in their ability to capture both the contextual information of large-scale objects and the fine details of small-scale objects. To further enhance the detection capability for multiscale objects, the DWA employs a multiscale feature aggregation strategy. This approach captures features at different levels of granularity, ensuring that both small and large objects are effectively represented.

Specifically, the module applies max-pooling operations with varying kernel sizes to generate multiscale features

$$\mathrm{MP}_i = \mathrm{MaxPool}_{k_i}(X_3), \quad k_i \in \{5, 9, 13\} \tag{14}$$

where k_i denotes the size of the pooling kernel and $\mathrm{MP}_i \in \mathbf{R}^{C\times H\times W}$ represents the pooled feature map at the corresponding scale. By using different kernel sizes, the module captures both fine and coarse contextual information, which is essential for detecting objects of varying sizes.

Next, the multiscale features are concatenated with the dynamically optimized feature map X_3 to form a comprehensive representation

$$X_{\mathrm{agg}} = \mathrm{Concat}(X_3, \mathrm{MP}_1, \mathrm{MP}_2, \mathrm{MP}_3) \tag{15}$$

where $X_{\mathrm{agg}} \in \mathbf{R}^{(C\times 4)\times H\times W}$ denotes the aggregated feature map. This concatenation ensures that information from different receptive fields is retained, promoting a more robust representation of objects at various scales.

Lastly, a 1×1 convolution is applied to fuse the multiscale features and produce the final output feature map

$$X_{\mathrm{out}} = \phi(X_{\mathrm{agg}}) \tag{16}$$

where $X_{\mathrm{out}} \in \mathbf{R}^{C\times H\times W}$ represents the final output. This mechanism ensures comprehensive coverage of both fine-grained details and broader contextual information, enabling the model to distinguish between densely distributed small objects and their background. By incorporating features from varying spatial extents, DWA significantly enhances the robustness and adaptability of the feature representation.

DWA is specifically designed to address the challenges of small-object detection in RSIs. By integrating context expansion, multiscale feature aggregation, and dynamic feature optimization, it mitigates feature confusion caused by densely arranged objects at both global and local levels. Consequently, the model's capability to accurately detect and distinguish small objects is significantly enhanced. This novel design offers robust support for the precise recognition of remote sensing objects in complex environments.

D. Prior-Guided Small-Object Detectors

Discrepancies in sensor distance and object size can cause small objects in RSIs to appear at different scales, with varying orientations depending on the imaging angle. These objects typically exhibit reduced texture and contrast, making them difficult to distinguish from the background or other objects using conventional models. Accurate detection of these objects requires a model capable of adjusting its receptive field across multiple scales while maintaining sensitivity to orientation changes. To address this challenge, we propose PGSOD, which incorporates PGCFM to enhance the model's ability to detect small objects through context-aware feature fusion and dynamic re-weighting, as shown in Fig. 6.

PGSOD consists of several stages designed to enhance feature representation at various scales while leveraging prior information to guide the detection process. The following part outlines how each component of PGSOD contributes to resolving the scale and orientation challenges inherent in small-object detection. The first key component is multiscale feature fusion, which allows the model to capture small objects at different scales. The process begins with upsampling lower-resolution feature maps, which are then concatenated with higher-resolution backbone features. This ensures the preservation of fine-grained features while simultaneously capturing larger contextual information. The feature fusion process starts with the upsampling of the feature map, which is then concatenated with feature maps from the C2f module of the backbone

$$\mathbf{X}_{\mathrm{up}} = \mathrm{Upsample}\big(\mathbf{X}_{\mathrm{input}}, \mathrm{scale} = 2\big) \tag{17}$$

$$\mathbf{X}_{\mathrm{concat}} = \mathrm{Concat}\big(\mathbf{X}_{\mathrm{up}}, \mathbf{P}_4\big). \tag{18}$$

Next, the concatenated feature map is processed using a series of C2f layers that enhance feature representation by incorporating residual connections for more effective context aggregation

$$\mathbf{X}_{C2f} = \mathrm{C2}f\,(\mathbf{X}_{\mathrm{concat}}, \mathrm{filters} = 512). \tag{19}$$

PGSOD then modifies the significance of each feature map channel based on its relevance to small-object detection, using a dynamic feature re-weighting mechanism informed by PGCFM. This mechanism dynamically emphasizes features most likely to correspond to objects while attenuating irrelevant background features

$$P_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F_{\mathrm{fusion}}(c, i, j). \tag{20}$$

The feature map is subjected to global pooling to extract channel-wise global features

$$w_c = \sigma(\mathbf{w_2} \cdot \delta(\mathbf{w_1} \cdot P_c)) \tag{21}$$

$$\mathbf{F}_{\text{re-weighting}} = \mathbf{F}_{\text{fusion}} \cdot w_c. \tag{22}$$

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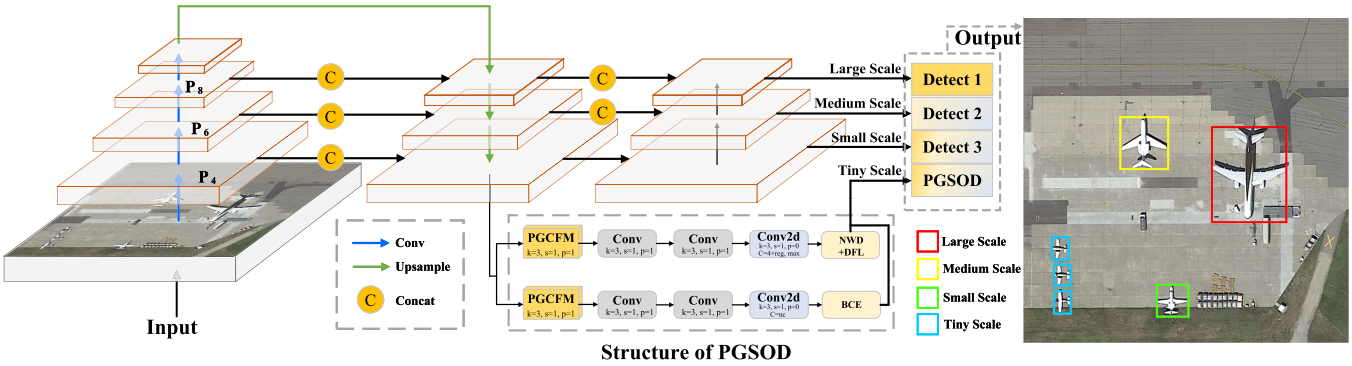


Fig. 6. Dashed box highlights the structure of PGSOD. To better illustrate the role of this small-object detection head, we have placed it within the simplified PGCFNet framework. We replace the CIOU loss function in the detection head with the NWD loss function, which more accurately measures the overlap between the object and prediction frames by calculating the distribution difference. This function demonstrates enhanced robustness, particularly when objects experience significant scale and orientation changes. By using NWD, we address scale and orientation challenges in RSIs more efficiently.

The global feature is passed through fully connected layers, followed by a sigmoid activation to produce dynamic channel weights. These weights are then applied to the feature map to re-weight the channels. This operation ensures that only the most relevant features are retained, improving the precision of object detection for small objects by eliminating background noise. Following feature re-weighting, the output undergoes a final 1×1 convolution to refine the feature map, ensuring it is compact and optimized for downstream detection tasks:

$$\mathbf{F}_{\text{output}} = \text{Conv}\, 1 \times 1\left(\mathbf{F}_{\text{re-weighting}}\right). \tag{23}$$

PGSOD addresses the challenges of small-object detection with a multistage feature fusion mechanism that incrementally combines features from various scales, ensuring effective capture of small objects regardless of transformations. This approach adaptively adjusts the importance of features across different spatial scales, improving the detection of objects with varying scales and orientations.

IV. Experiments

This section begins by describing the publicly available datasets, each of which presents significant challenges. Next, the evaluation metrics and implementation details for all experiments are provided. Finally, the effectiveness of each proposed module is demonstrated through an ablation study. The results, including those for detection, feature extraction, and comparisons with SOTA methods, are presented in subsequent sections. Additionally, we validate the effectiveness of the proposed PGCFM by integrating it into YOLOv9 [29] and YOLOv10 [30] across various network sizes and datasets.

A. Experimental Datasets Description

1) RSOD [14]: The RSOD dataset, proposed by Wuhan University, is a remote sensing dataset that includes categories such as aircraft, oil tanks, overpasses, and playgrounds. It contains 976 images and 6950 object instances, including 4493 aircraft, 1586 oil tanks, 191 playgrounds, and 180 overpasses. The image sizes range from 512×512 to 1961×1193 pixels, with spatial resolutions varying between 0.3 and 3 m. We use 75% of the images for training and 25% for validation.

2) DIOR [12]: The DIOR dataset serves as a comprehensive benchmark for object detection in optical RSIs. It consists of 23 463 images and 192 472 object instances across 20 categories: airplane (AL), airport (AT), baseball field (BF), basketball court (BC), bridge (BG), chimney (CM), dam (DM), expressway service area (EA), expressway toll station (ES), vehicle (VH), harbor (HB), overpass (OP), golf course (GC), ground track field (GF), ship (SP), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), and windmill (WM). The spatial resolution of the dataset ranges from 0.5 to 30 m. The training set consists of 5862 images, while the validation and test sets contain 5863 and 11 738 images, respectively.

3) NWPU VHR-10 [13]: The NWPU VHR-10 dataset, introduced by Northwestern Polytechnic University in 2014, is widely used in remote sensing applications. The latest version contains 1172 images categorized into ten classes: airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor port, bridge, and vehicle. These ten categories are further divided into positive and negative sets. The spatial resolution of the images ranges from 0.08 to 2 m.

4) AI-TOD [15]: AI-TOD is a specialized dataset designed for tiny object detection in aerial images. Unlike other RSOD datasets, AI-TOD features an average object size of approximately 12.8 pixels, significantly smaller than those in existing datasets. It consists of 28 036 aerial images, containing a total of 700 621 object instances. These instances are categorized into eight classes, including airplanes, bridges, and storage tanks. For model evaluation, the dataset provides a training set of 11 214 images, a validation set of 2804 images, and a test set of 14 018 images, in accordance with the official guidelines.

B. Evaluation Metrics

The evaluation metrics used are the average precision (AP) for each class and the mean average precision (mAP) across all classes, defined as follows:

$$\text{AP} = \int_0^1 P_r(R_e) dR_e \tag{24}$$

$$\text{mAP} = \frac{1}{N_{\text{cls}}} \sum_{i=1}^{N_{\text{cls}}} AP_i \tag{25}$$

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TABLE I
TRAINING ENVIRONMENT AND HARDWARE PLATFORM PARAMETERS

Hardware	Configuration
CPU	i9-12900K
GPU	NVIDIA GeForce RTX 3090
GPU memory size	24G
Operating systems	Win10
Deep learning architecture	Pytorch1.12.1 + Cuda11.3 + cudnn11.3

TABLE II
KEY PARAMETERS SET DURING MODEL TRAINING

Parameters	Setup
epochs	600
Momentum	0.932
Initial learning rate	0.01
Final learning rate	0.01
Weight decay	0.0005
Batch size	4
Input image size	640×640
Optimizer	SGD

where P_r and R_e denote the precision and the recall, respectively, and N_{cls} serves as the total number of classes. The precision P_r and the recall R_e can be expressed as follows:

$$P_r = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{26}$$

$$R_e = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{27}$$

where N_{TP}, N_{FP}, and N_{FN} denote the amount of the true positives (TP_s), false positives (FP_s), and false negatives (FN_s), respectively. In particular, TP refers to data instances that are both correctly identified and accurately classified. Conversely, FP denotes instances that are either incorrectly detected or misclassified. Additionally, FN describes instances that fail to be detected. A prediction qualifies as TP if its intersection over union (IoU) with the ground truth exceeds 0.5 during the inference; otherwise, it is classified as FP.

C. Implementation Details

The experiments were conducted using the PyTorch 1.6 environment on a Windows 10 system, with an NVIDIA RTX 3090 GPU (24 GB of RAM) as detailed in Table I. Detection results were evaluated using the mAP at an IoU threshold of 0.5. The hardware and environmental parameters used in the training phase are listed in Table II.

D. Ablation Study and Analysis

To assess the contribution of each component in PGCFNet, we systematically incorporate the PGCFM, PGSOD, and DWA modules into the baseline model and evaluate their effectiveness. The ablation experiment was conducted on the RSOD dataset. Table III presents the impact of adding or removing each module on the evaluation metrics. To further demonstrate the effectiveness of our proposed PGCFNet, we visualize its prediction results on RSOD, as shown in Fig. 7 presents the detection results for four typical categories in the RSOD dataset. The results clearly demonstrate that PGCFNet achieves exceptional performance and robustness across various object categories. Notably, for the detection of airplanes with significant directional changes, PGCFNet accurately identifies objects from multiple orientations, showcasing its adaptability in complex scenarios. Additionally, in the densely arranged oil tank scene, PGCFNet effectively mitigates feature confusion between adjacent objects, ensuring precise detection even in high-density environments. These findings further confirm the superiority of PGCFNet for multiscale object detection in diverse and challenging remote sensing imagery.

1) Effect of PGCFM: As shown in the second row of Table III, the addition of PGCFM leads to a noticeable improvement in performance. Specifically, when PGCFM is incorporated alone, we observe an increase in mAP_{50} from 93.4% to 94.6% and a corresponding improvement in recall from 88.5% to 90.5%. This indicates that PGCFM enhances the model's ability to detect objects more accurately, particularly in challenging remote sensing environments where weak textures and background noise may obscure object features.

2) Effect of DWA: The results in the third row of Table III show that DWA leads to an increase in mAP_{50} from 93.4% to 94.3%, as well as a substantial boost in recall to 93.8%. This demonstrates that DWA enables the model to dynamically assess the importance of spatial regions across feature maps at different scales, assigning higher weights to critical areas. Consequently, this facilitates precise spatial alignment and enhances overall detection accuracy.

3) Effect of PGSOD: The fourth row of Table III illustrates that incorporating PGSOD results in a moderate improvement in mAP_{50}, which increases to 94.8%, while recall reaches 92.0%. These findings suggest that PGSOD helps the model focus more effectively on critical object regions, particularly enhancing detection accuracy for small and overlapping objects.

As illustrated in the last row of Table III, the model incorporating all three modules (PGCFM, DWA, and PGSOD) demonstrates optimal performance, with mAP50 reaching 95.8%, mAP50 : 95 at 67.7%, and recall at 92.0%. These results validate the efficacy of integrating these modules for achieving superior detection performance in RSOD. The combination of PGCFM, DWA, and PGSOD effectively suppresses background noise, reduces feature confusion between adjacent objects, and optimizes focus on critical regions, respectively. To further assess the impact of PGCFM, DWA, and PGSOD on detection performance, we visualized the feature maps for airplanes, oil tanks, ships, and vehicles, as shown in Fig. 8. As seen in Fig. 8(b), the raw feature maps capture basic object features but also contain considerable background information and noise, such as sea texture and wave patterns, which hinder small-object detection.

Furthermore, the features of the oil tanks are intermixed, making it difficult to accurately distinguish individual objects. Fig. 8(c) shows that applying PGCFM effectively suppresses background noise and distracting features. Fig. 8(d) illustrates the improvements achieved by combining PGCFM and DWA, resulting in more distinct and well-separated object features.

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(a) (b) (c) (d)

Fig. 7. Visualizing the detection results of our PGCFNet on the RSOD dataset. (a) Airplane. (b) Overpass. (c) Oil tank. (d) Playground.

TABLE III
ABLATION EXPERIMENTS FOR PGCFM, DWA, AND PGSOD IN RSOD

PGCFM	DWA	PGSOD	$mAP_{50}(\%)$	$mAP_{50:95}(\%)$	Recall(%)
			93.4	66.1	88.5
✓			94.6	66.4	90.5
	✓		94.3	67.4	**93.8**
		✓	94.8	66.8	92.0
✓		✓	95.0	66.2	91.6
✓	✓		95.5	67.5	93.1
✓	✓	✓	**95.8**	**67.7**	92.0

Despite the dense arrangement of oil tanks and vehicles, DWA significantly reduces feature confusion between adjacent objects. Finally, Fig. 8(e) presents the results after processing with PGCFM and DWA, followed by optimization through PGSOD. The previously dispersed attention is now focused on key object regions, leading to substantial improvements in detection accuracy and reliability.

E. Comparison Experiments and Analysis

This section presents a comparative analysis of PGCFNet against other SOTA detectors using the DIOR, NWPU VHR-10, and AI-TOD datasets. Results are detailed in Tables IV–VI.

1) Results on DIOR: We evaluate our proposed PGCFNet on the DIOR dataset, as presented in Table IV, comparing it with several SOTA methods: Faster R-CNN [31], CF2PN [32], Cascade R-CNN [33], CSFFNet [34], SRAFNet [35], ASSD [36], RSADet [37], SB-MSN [38], TMAFNet [39], AGMFNet [40], LGDA [41], CoF-Net [26], GLSANet [42], ASFANet [43], and YOLOv8 m [28]. As shown in Table IV, PGCFNet achieves the highest mAP, outperforming competing methods in individual categories. Specifically, PGCFNet achieves an mAP exceeding 86.0%, surpassing the second-best result by 2.7%, demonstrating a significant improvement in accuracy on the DIOR dataset. In terms of category-wise performance, PGCFNet consistently demonstrates superior precision, ranking first in 11 of the 20 categories. Notably, PGCFNet achieves precision above 90.0% in 11 categories, significantly outperforming other methods. In particular, for categories such as BG and ES, PGCFNet's precision exceeds that of the second-best method by 3.8% and 3.2%, respectively. Even in more challenging categories, such as BG, where precision is relatively low at 59.0%, PGCFNet maintains a notable advantage, surpassing the nearest competitor (AFSNet) by 2.8%.

Fig. 9 presents the detection results of PGCFNet on the DIOR dataset for several representative scenarios. Despite the presence of large and complex backgrounds in the RSIs, our method effectively detects objects, particularly the windmill in Fig. 9(e) and the ground track field in Fig. 9(h). These results underscore the effectiveness of PGCFM within PGCFNet. By addressing key challenges, such as weak textures in RSIs, PGCFM enhances the network's capacity to extract and integrate critical contextual information, leading to significant improvements in both overall accuracy and category-specific precision. The visual results further validate the superior performance of PGCFM, DWA, and PGSOD in multiscale object detection within RSIs.

2) Results on NWPU VHR-10: Table V compares the performance of PGCFNet with other SOTA detectors on the NWPU VHR-10 dataset. The competing methods include HRBM [44], YOLOv3 [45], Faster R-CNN [31], FPN [46], EGAT [47], YOLOv7 [48], YOLOv8 [28], LGDA [41], CoF-Net [26], and GLSANet [42]. In general, PGCFNet exhibits exceptional precision, achieving an mAP of 95.92%, the highest among all evaluated models. Specifically, PGCFNet achieves an AP of 99.12% for BCs and 97.40% for vehicles. Notably, the vehicle images consist of locally sharpened color-infrared images rather than conventional satellite RSIs. Despite this difference, PGCFNet achieves the highest accuracy (97.40%), outperforming the second-best method, CoF-Net, by over 6.80%. These results underscore the superior adaptability and robustness of PGCFNet compared to other competing methods.

3) Results on AI-TOD: Table VI presents a performance comparison between our PGCFNet and other SOTA detectors on the AI-TOD dataset. The competitive methods include

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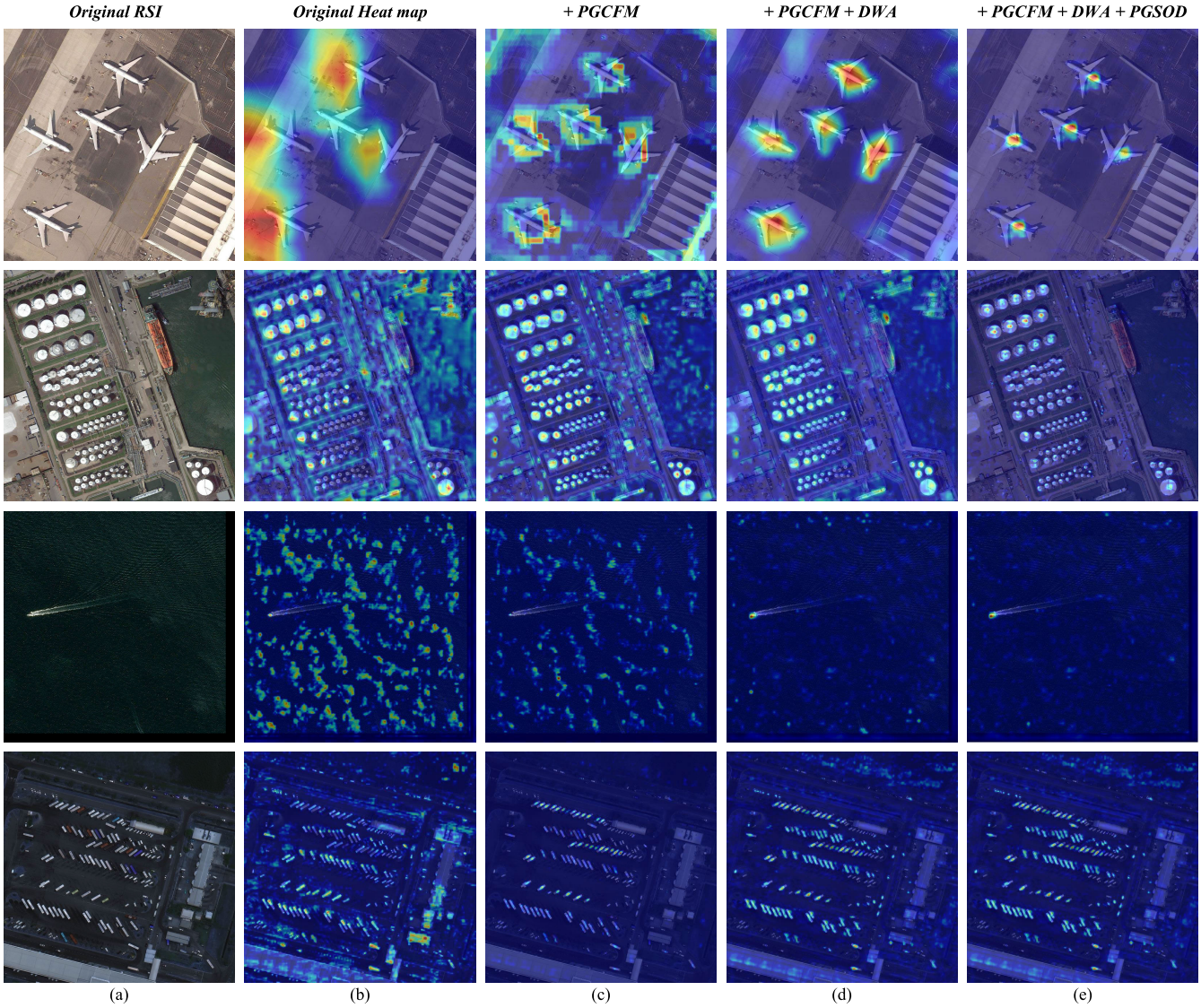


Fig. 8. Influence of PGCFM, DWA, and PGSOD on feature extraction. We selected four challenging RSIs to validate the proposed modules, which, respectively, represent orientation changes, scale variations, weak texture, and densely arranged small objects. The first row illustrates airplanes, which exhibit substantial variations in orientation among objects. The second row displays oil tanks, characterized by their dense arrangement as small objects. The third and fourth rows show ships and vehicles, respectively. Ships are influenced by the textured background of ocean waves, whereas vehicles encounter a combination of all these challenges.

TABLE IV

COMPARISON WITH SOTA ON DIOR [12]. THE BEST RESULTS ARE MARKED IN BOLD

Method	mAP(%)	AL	AT	BF	BC	BG	CM	DM	EA	ES	GC	GF	HB	OP	SP	SD	ST	TC	TS	VH	WM
Faster R-CNN [31]	66.9	81.1	76.9	89.8	80.5	45.1	81.7	63.4	80.9	63.2	73.3	78.3	43.3	58.4	68.5	89.1	59.0	81.1	60.2	47.8	79.3
CF2PN [32]	67.3	78.3	78.3	76.5	88.4	37.0	71.0	59.9	71.2	51.2	75.6	77.1	56.8	58.7	76.1	70.6	55.5	88.8	50.8	36.9	86.4
Cascade R-CNN [33]	67.4	81.2	81.4	90.1	81.1	46.9	81.5	68.1	84.2	65.3	74.5	81.6	37.8	60.4	68.9	88.8	60.5	81.0	57.5	47.7	80.7
CSFFNet [34]	68.0	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	81.6	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9
SRAF-Net [35]	69.7	88.4	76.5	92.6	87.9	35.8	83.8	58.6	86.8	66.8	76.4	82.8	16.2	58.0	59.4	80.9	55.6	90.6	52.0	53.2	91.0
ASSD [36]	71.1	85.6	82.4	75.8	89.5	40.7	77.6	64.7	67.1	61.7	80.8	78.6	62.0	58.0	84.9	76.7	65.3	87.9	62.4	44.5	76.3
RSADet [37]	72.2	73.6	86.0	72.6	89.6	43.6	75.3	62.3	79.5	68.7	78.6	79.1	57.9	59.2	90.0	55.8	77.0	87.8	65.3	55.3	86.5
SB-MSN [38]	70.3	79.6	82.2	76.4	89.8	45.6	78.2	64.8	58.9	59.3	79.2	82.4	51.8	60.8	74.4	79.7	66.4	85.6	65.4	55.3	86.5
TMAFNet [39]	72.9	92.2	77.7	75.0	91.3	47.1	78.6	53.6	67.1	66.2	78.5	76.3	64.9	61.4	90.5	72.2	75.4	90.7	62.1	55.2	83.2
AGMF-Net [40]	73.2	90.9	72.8	79.3	89.7	44.7	81.4	59.3	66.0	62.7	73.8	79.2	65.0	61.7	91.7	78.6	75.8	90.7	60.0	58.0	83.1
LGDA [41]	74.1	79.6	83.4	81.8	89.2	52.8	73.7	68.1	84.5	61.2	77.3	81.9	60.1	55.2	89.7	64.9	82.7	85.2	63.9	58.5	88.6
CoF-Net [26]	75.8	84.0	85.3	82.6	90.0	47.1	80.7	73.3	89.3	74.0	84.5	83.2	57.4	62.2	82.9	77.6	68.2	89.9	68.7	49.3	85.2
GLSANet [42]	77.9	95.8	78.9	92.9	87.9	50.7	81.1	55.5	79.8	74.1	71.6	87.6	66.4	65.5	95.2	92.4	86.3	94.8	50.6	62.1	89.2
AFSNet [43]	78.9	91.6	83.4	**96.1**	**94.9**	56.2	87.6	**79.4**	64.6	76.8	**87.9**	73.3	60.5	61.7	94.3	87.4	77.0	89.2	65.7	57.9	92.9
YOLOv8m [28]	83.3	96.1	89.4	92.4	89.2	53.4	88.0	74.3	**95.9**	**80.0**	84.4	89.4	71.9	74.2	93.7	**95.6**	86.2	92.0	**70.8**	**72.9**	94.2
PGCFNet (ours)	**86.0**	**97.7**	**90.7**	94.3	91.0	**59.0**	**90.0**	76.3	94.6	79.6	87.4	**91.0**	**74.4**	**77.9**	**96.3**	95.0	**89.3**	**97.0**	62.2	70.0	**95.7**

M-CenterNet [15], Cascade R-CNN [33], DetectoRs [49], DerectoRS + NWD [50], SP-YOLOv8s [51], HANet [52], and L-FFCA-YOLO [53]. On the AI-TOD dataset, PGCFNet achieves the highest performance, with a mAP_{50} of 58.5% and

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Fig. 9. Detection results of PGCFNet in DIOR for representative scenarios. (a) and (b) Airplane. (c) Baseball field, basketball court, and vehicle. (d) Oil tank. (e) Windmill. (f) Train station and bridge. (g) Ship, tennis court, and vehicle. (h) Ground track field and stadium. (i) Ship. (j) Chimney. (k) Expressway toll station. (l) Dam.



Fig. 10. Visualizing the detection results of our PGCFNet on the AI-TOD and NWPU VHR-10. (a) and (d) Ship. (b) and (e) Vehicle and baseball court. (c) and (f) Tennis court, ground track field, baseball diamond, and baseball court.

an $mAP_{50:95}$ of 27.6%. It surpasses the second-best method, L-FFCA-YOLO, by 0.2% in mAP_{50} and 2.1% in $mAP_{50:95}$. This improvement highlights the effectiveness of PGCFNet's prior-guided context fusion and multiscale feature extraction capabilities. These advancements allow PGCFNet to handle challenges such as the dense arrangement of small objects

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TABLE V
COMPARISON EXPERIMENTS FOR PGCFNET IN NWPU VHR-10 [13]. THE BEST RESULTS ARE MARKED IN BOLD

Method	mAP	Airplane	Ship	ST	BD	TC	BC	GT	Harbor	Bridge	Vehicle
HRBM [44]	87.12	99.70	90.80	90.61	92.91	90.29	80.13	90.81	80.26	68.53	87.14
YOLOv3 [45]	87.27	99.55	81.28	80.30	98.26	80.56	81.82	99.47	74.13	89.61	86.98
Faster R-CNN [31]	88.30	**100**	85.28	**100**	95.93	87.59	92.08	99.73	92.11	43.37	80.60
FPN [46]	90.80	**100**	90.86	99.99	96.84	90.67	95.05	**100**	93.67	50.86	90.10
EGAT [47]	92.01	97.32	90.86	97.16	96.51	86.64	94.46	94.18	86.18	80.14	90.79
YOLOv7 [48]	92.94	99.52	**96.72**	96.61	97.78	93.85	90.78	95.60	89.54	83.72	89.38
YOLOv8 [28]	93.61	99.89	92.60	96.67	97.93	93.90	90.82	96.79	90.65	84.69	90.60
LGDA [41]	94.08	98.75	91.28	95.70	98.43	96.93	94.62	94.17	93.05	89.81	88.02
CoF-Net [26]	94.51	**100**	90.62	96.39	98.85	91.54	95.73	99.70	92.05	89.58	90.60
GLSANet [42]	94.57	99.90	94.01	97.12	**99.40**	98.75	86.12	99.46	97.51	84.83	86.76
PGCFNet (ours)	**95.92**	99.50	91.53	99.38	97.03	98.20	**99.12**	98.77	93.91	84.36	**97.40**

TABLE VI
COMPARISON EXPERIMENTS FOR PGCFNET IN AT-TOD

Methods	mAP_{50}	$mAP_{50:95}$
M-CenterNet [15]	0.407	0.145
Cascade R-CNN [33]	0.308	0.138
DetectoRS [49]	0.328	0.148
DetectoRS + NWD [50]	0.493	0.208
SP-YOLOv8s [51]	0.483	0.227
HANet [52]	0.537	0.221
L-FFCA-YOLO [53]	0.583	0.255
PGCFNet (ours)	**0.585**	**0.276**

TABLE VII
GENERALITY EXPERIMENTS FOR PGCFM ON RSOD AND DIOR

Method	Model Size	Epoch	Dataset	Without PGCFM	With PGCFM
YOLOv9 [29]	c	50	DIOR	79.1	**79.7**
YOLOv9 [29]	s	50	DIOR	72.5	**74.0**
YOLOv9 [29]	s	100	RSOD	90.0	**90.2**
YOLOv10 [30]	s	-	RSOD	92.5	**92.7**
YOLOv10 [30]	n	-	RSOD	91.2	**92.1**
YOLOv10 [30]	l	-	RSOD	92.0	**94.3**
YOLOv10 [30]	m	-	DIOR	85.5	**85.8**

TABLE VIII
GENERALITY EXPERIMENTS FOR DWA ON RSOD AND DIOR

Method	Model Size	Epoch	Dataset	Without DWA	With DWA
YOLOv9 [29]	c	50	DIOR	79.1	**80.5**
YOLOv9 [29]	s	50	DIOR	72.5	**73.0**
YOLOv9 [29]	s	100	RSOD	**90.0**	89.1
YOLOv10 [30]	s	-	RSOD	92.5	**92.8**
YOLOv10 [30]	n	-	RSOD	91.2	**91.4**
YOLOv10 [30]	l	-	RSOD	92.0	**92.6**
YOLOv10 [30]	m	-	DIOR	**85.5**	84.3

and significant scale variability. Compared to HANet and SP-YOLOv8s, PGCFNet exhibits enhanced adaptability and precision, particularly in scenarios involving complex backgrounds and low-contrast objects.

We further compare PGCFNet with both advanced and SOTA methods using the AI-TOD and NWPU VHR-10 datasets. Fig. 10 illustrates the detection results of PGCFNet in representative scenarios across these diverse datasets. The results show that PGCFNet excels in detecting multiscale objects, even in complex backgrounds, emphasizing its robustness and adaptability. The model effectively distinguishes small objects from cluttered environments and adapts to variations in object scale, orientation, and density. These findings highlight the strong performance of PGCFNet in RSOD, particularly under challenging conditions.

F. Generality Experiment

The experimental results demonstrate the generalization capability of PGCFM, DWA, and PGSOD across multiple YOLO versions and datasets, as shown in Table VII. For YOLOv9, integrating PGCFM resulted in a significant performance improvement. The small model on the DIOR dataset showed a 1.5% increase in mAP (from 72.5% to 74.0%). Similarly, the compact YOLOv9 model on DIOR achieved a 0.6% improvement (from 79.1% to 79.7%). On the RSOD dataset, PGCFM led to a modest 0.2% improvement in the small YOLOv9 model after 100 epochs (from 90.0% to 90.2%).

For YOLOv10, PGCFM further enhanced detection performance across various model sizes. The small model showed a 0.2% improvement (from 92.5% to 92.7%), while the large model exhibited a more substantial gain of 2.3% (from 92.0% to 94.3%) on the RSOD dataset. Even on the DIOR dataset, the medium-sized YOLOv10 model achieved a slight 0.3% increase in mAP (from 85.5% to 85.8%) with PGCFM.

Furthermore, the generalization capability of DWA was evident in both YOLOv9 and YOLOv10 as shown in Table VIII. On the DIOR dataset, incorporating DWA into YOLOv9 led to an increase of 1.1% in mAP for the small model (from 71.2% to 72.3%). Similarly, YOLOv10 with DWA achieved a notable gain of 1.8% in mAP on the RSOD dataset for the large model (from 91.5% to 93.3%).

PGSOD also demonstrated effective generalization across different datasets and model sizes as shown in Table IX. In YOLOv9, integrating PGSOD improved the mAP by 0.9% on the DIOR dataset with the medium model (from 78.3% to 79.2%). On the RSOD dataset, the large YOLOv10 model showed a performance increase of 1.5% (from 89.8% to 91.3%). These results confirm that PGCFM, DWA, and

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TABLE IX
GENERALITY EXPERIMENTS FOR PGSOD ON RSOD AND DIOR

Method	Model Size	Epoch	Dataset	Without PGSOD	With PGSOD
YOLOv9 [29]	c	50	DIOR	79.1	**81.3**
YOLOv9 [29]	s	50	DIOR	72.5	**73.7**
YOLOv9 [29]	s	100	RSOD	90.0	**91.4**
YOLOv10 [30]	s	-	RSOD	92.5	**93.7**
YOLOv10 [30]	n	-	RSOD	91.2	**92.0**
YOLOv10 [30]	l	-	RSOD	92.0	**92.8**
YOLOv10 [30]	m	-	DIOR	85.5	**86.5**

PGSOD consistently improve performance across different model sizes and datasets, particularly in detecting small and challenging objects.

V. CONCLUSION

In this article, we propose a novel object detection network, PGCFNet, aimed at enhancing small-object detection in RSIs. The design of PGCFNet focuses on effectively aggregating prior contextual information to strengthen feature representation. First, we introduce PGCFM, which integrates boundary and background information through dynamic weighted fusion and prior knowledge, minimizing background interference. Second, we employ DWA to adaptively balance multiscale features. Finally, PGSOD is developed to address challenges related to scale and orientation variations in small objects. Extensive experiments conducted on three public datasets demonstrate the superior performance of PGCFNet, particularly in overcoming challenges such as weak textures, scale variations, and densely arranged objects.

Additionally, generalization experiments of PGCFM, DWA, and PGSOD validate their adaptability across different datasets and detection networks. This study makes a significant contribution to small-object detection in RSIs, especially by effectively leveraging prior knowledge. However, the study also presents some limitations, including increased experimental complexity due to the need for adjusting loss function weights across different datasets, as well as higher computational demands. Future work will focus on optimizing the model's computational efficiency and further improving its performance in real-world applications.

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